



A DNA assembly model of sentence generation

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ABSTRACT

Recent results of corpus-based linguistics demonstrate that context-appropriate sentences can be generated by a stochastic constraint satisfaction process. Exploiting the similarity of constraint satisfaction and DNA self-assembly, we explore a DNA assembly model of sentence generation. The words and phrases in a language corpus are encoded as DNA molecules to build a language model of the corpus. Given a seed word, the new sentences are constructed by a parallel DNA assembly process based on the probability distribution of the word and phrase molecules. Here, we present our DNA code word design and report on successful demonstration of their feasibility in wet DNA experiments of a small scale.

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1. Introduction

DNA molecules have been used for various purposes. Much of their use exploits the self-assembly property of DNA (Adleman, 1994; Mao et al., 2000; Maune et al., 2010; Paun et al., 1998; Reif and LaBean, 2007; Seeman, 2003; Winfree et al., 1998; Yan et al., 2003; Zheng et al., 2009). The nucleotides of A, T, G, and C recognize their counterparts by Watson–Crick complementarity (A–T and G–C) and this molecular recognition property can be used to assemble small DNA-molecular structures into larger structures. Many information processes can be formulated as a DNA self-assembly process once the molecular building blocks are defined and their target structures appropriately represented. In this paper we explore the potential of using DNA self-assembly for sentence generation based on a language corpus. Recent studies in usage-based linguistics (Tomasello, 2003; Tummers et al., 2005) and corpus-based language processing (Biber et al., 1998; Foster, 2007; Heylen et al., 2008; McEnery et al., 2006) show that language models can be built

from a language corpus, and sentences can be generated by performing a stochastic constraint satisfaction process, i.e. satisfying the conditions or constraints encoded by micro-rules or features extracted from the language corpus (Rosenfeld et al., 2001).

One of the cores of human linguistic capabilities is to assemble words into the proper sequence, decompose sentences into small fragments and words, and reconstruct sentences with phrases and words. Once the words are encoded as DNA molecules we may exploit the molecular recognition properties of DNA to assemble the words into phrases and sentences. By designing the molecules to reflect the constraints expressed in the language corpus, the language generation can be naturally simulated by the DNA assembly process. Recently, simulation studies were conducted to evaluate this potential (Lee et al., 2009; Zhang and Park, 2008). They collected drama dialogues to build a language corpus and used a specialized probabilistic graphical model called a hypernetwork to build a language model. Once the hypernetwork model is constructed, new sentences are generated from the model by assembling the linguistic fragments that match with the given seed words. Since the encoding and decoding of sentences are modeled after the DNA hypernetworks (Zhang and Kim, 2006), the whole process can be implemented in DNA wet experiments.

In this paper we present a method for encoding the DNA sequences to represent the words and phrases. We also present a DNA-based model for assembling the words into phrases to generate sentences. The procedures are verified by *in vitro* DNA experiments on a vocabulary of 8 words. The results are confirmed

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by sequencing the DNA products resulting from the assembly processes. We also illustrate and demonstrate the potential of the DNA assembly procedures for language processing by performing *in silico* simulations on a large corpus of sentences collected from TV dramas.

2. The DNA assembly model of language

The language model is constructed using the DNA hypernetwork structure (Zhang, 2008). The language hypernetwork consists of a large number of hyperedges which correspond to phrases, where each hyperedge consists of words constituting the phrase. Each word can be encoded as a DNA sequence and each phrase or hyperedge as a consecutive sequence of DNA molecules. Given a corpus of sentences, a language hypernetwork is built by sampling many hyperedges from each sentence. In generating a new sentence given a query word, the hyperedges are assembled that match with the given query word. Experimentally, the encoded DNA hyperedges are collected and mixed in a microtube, and DNA sentences are generated by DNA self-hybridization process. The relevant sentences are filtered. The filtered DNA sentences are read by DNA sequencing. More detailed procedures and encoding schemes are described below. For illustration and verification, we implement small scale DNA experiments that generate sentences using 8 words.

2.1. Language hypernetwork model

A hypernetwork is defined as a triple $H=(X,E,W)$ where $X=\{x_1, x_2, \dots, x_i\}$, $E=\{E_1, E_2, \dots, E_i\}$, and $W=\{w_1, w_2, \dots, w_i\}$ are the sets of vertices, hyperedges, and weights, respectively (Zhang, 2008). Fig. 1 shows a language hypernetwork model for sentence generation. The vertex represents a word and the hyperedge represents a group of words or a phrase. The weight of the hyperedge represents the frequency of the phrase or its strength.

The number of words in a hyperedge is defined as its order. Two different methods are possible for extracting the hyperedges from the corpus: random extraction and sequential extraction. Random extraction samples the phrases from arbitrary positions of the sentence, while sequential extraction samples the phrases from consecutive positions of the sentence. In this paper we describe the sequential extraction method, however the method can be easily modified to perform random extraction. The sequential extraction method works by moving the order-size window to the right in a word-by-word manner to extract hyperedges. For example, given the sentence ‘The airport has only one runway’, four hyperedges of order-three are extracted from it, generating ‘The airport has’, ‘airport has only’, ‘has only one’, and ‘only one runway’. In this way, the hyperedges extracted from the corpus are connected, making a hypernetwork structure.

Once the hypernetwork is constructed, new sentences are generated by the following procedure. In this setup, we assume that the query (x_q) is given as a single word. The new words are then extended bidirectionally, i.e. left and right simultaneously.

Step 1. Given a keyword $X_q=(x_q)$, retrieve the hyperedges into $M=\{E_1, E_2, \dots, E_m\}$.

Step 2. Select a hyperedge $E_q=(x_{q-1}, x_q, x_{q+1})$ from M using roulette wheel selection.

Step 3. Set $X_q=(x_{q-1})$ and do Steps 1 and 2. Set the resulting hyperedge $E_{left}=(x_{q-2}, x_{q-1}, x_q)$.

Step 4. Set $X_q=(x_{q+1})$ and do Steps 1 and 2. Set the resulting hyperedge $E_{right}=(x_q, x_{q+1}, x_{q+2})$.

Step 5. Generate a (partial) sentence $S_q=(x_{q-2}, x_{q-1}, x_q, x_{q+1}, x_{q+2})$ by concatenating E_{left} and E_{right} .

(a) $H=(X, E, W)$

$X=(\text{good, friend, best, my, a, have, your})$

$E=(\text{my good friend, my best friend, have a friend, your best friend, a good friend})$

$W=(3, 2, 1, 4, 1)$

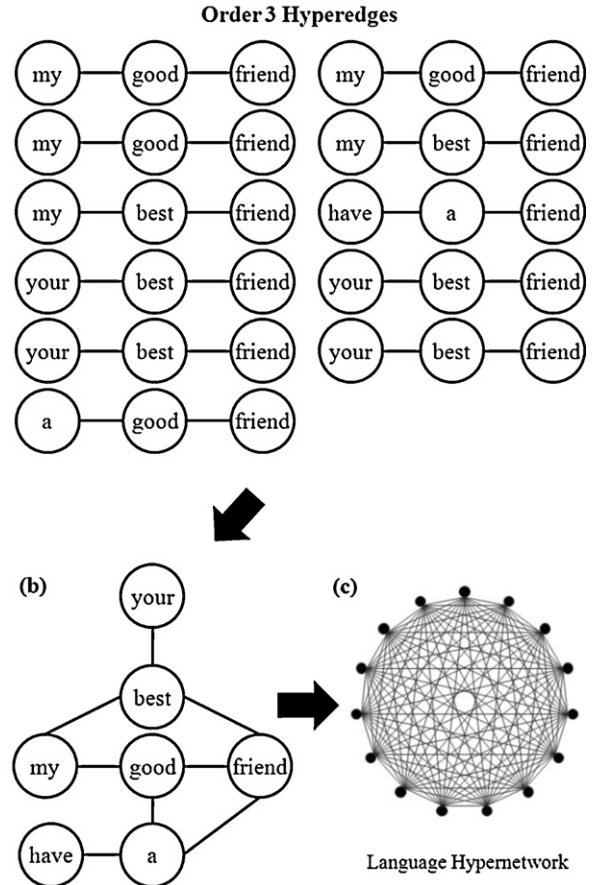


Fig. 1. A language hypernetwork model. (a) Hypernetwork H consists of a set X of vertices representing the words, a set E of hyperedges representing the phrases (hyperedges are edges that can connect an arbitrary number of vertices), a weight set W of the hyperedges. (b) The linked structure of the order 3 hyperedges. (c) The hypernetwork model of the sentences, constructed from the entire corpus of sentences.

Step 6. Repeat Steps 3–5 for $X_q=(x_{q-2})$ and $X_q=(x_{q+2})$ until the termination condition is met.

The procedure terminates if the candidate words belong to the terminal words (the first and last words in the sentences) with a high probability. The ‘generation by DNA assembly’ procedure is illustrated in Fig. 2. It shows how to generate a sentence ‘I’ll talk to my friend and my sister’ from the keyword, ‘friend’. Table 1 shows the example sentences generated by the bidirectional assembly process. In this simulation, a hypernetwork of order $k=3$ hyperedges was learned on the corpus of TV dramas *Friends*. The corpus consists of 63,152 sentences composed of 385,285 words. The left column is the result for query word ‘friend’ and the right column is for ‘you’.

We find that most of the sentences generated are novel (i.e. not in the original corpus) and make sense. Considering the size of the vocabulary (approximately 15,000 for *Friends*), the theoretical number of potential sentences is enormously large. We note that the sentence assembly process should satisfy the constraints that neighboring words to those already attached are the ones that fit the most natural context. In this regard, it has a similarity to the

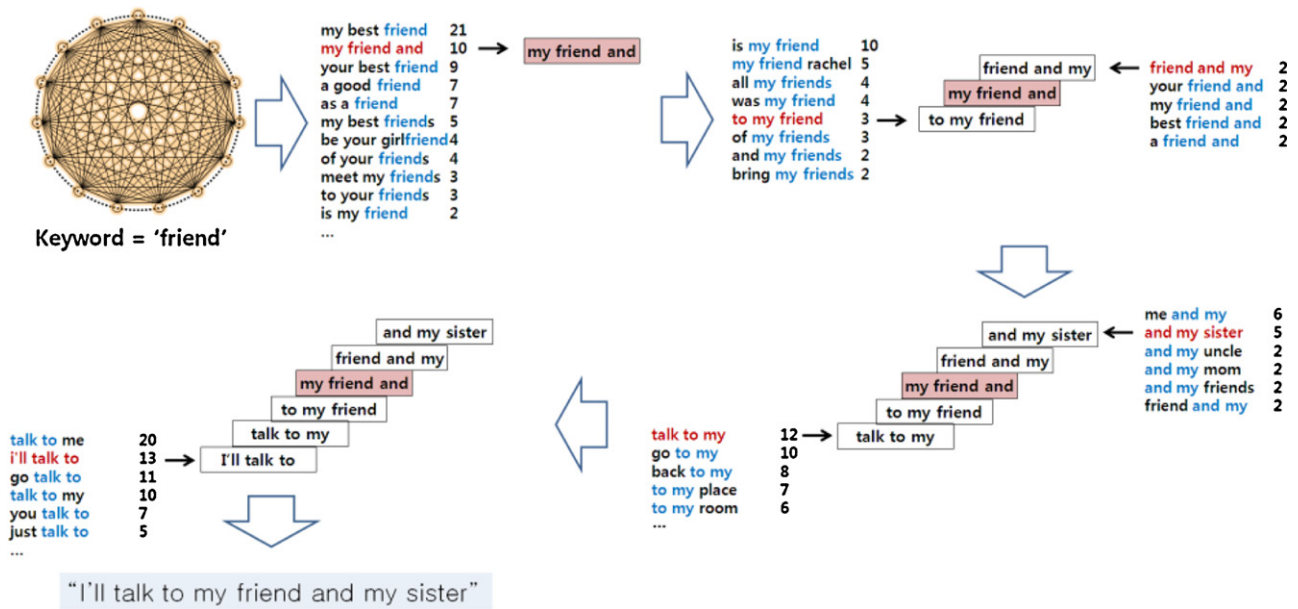


Fig. 2. Generating a new sentence from a keyword (in this case, "friend"). The given keyword is extended by assembling new words to the left and right ends of the existing partial sentence. The procedure illustrates the iterative process of composing from "friend" to "my friend and" through "to my friend and my", "talk to my friend and my sister", and finally to "I'll talk to my friend and my sister".

constraint satisfaction process underlying the maximum entropy language model in traditional language modeling (Rosenfeld et al., 2001).

2.2. Encoding

The hyperedges extracted from sentences to be learned are encoded as DNA hyperedges. Fig. 3 shows the encoding scheme that we have specifically designed for this purpose.

A DNA sequence representing a word in a sentence is referred to as a DNA word. Each word of a sentence is substituted for a distinct DNA word. (for example, I'll = A(a), talk = B(b), ..., the upper and lower cases are complementary DNA sequences of each other.) The number of hyperedges used for DNA experiments is six as shown

in Fig. 3. In the encoding process, the DNA word in the center of each hyperedge is repeated twice. The center DNA word is duplicated to connect the DNA hyperedges bidirectionally to each other *in vitro*. If the central DNA word is not repeated, only one DNA word is matched in one direction of the hyperedge as shown in Fig. 4(a), and chances are higher that nonsense DNA sentences are generated. The upper and lower cases representing each DNA word are complementary DNA sequences, and these are hybridized to each other, forming double-strand DNA molecules. The hyperedges, marked as

Table 1

Sentences generated for different keywords, 'friend' (left column) and 'you' (right column), in the same drama *Friends*.

Corpus: <i>Friends</i> Keyword: "friend"	Corpus: <i>Friends</i> Keyword: "you"
This is my friend rachel	You don't have that
My good friend	I thought you were in your life
You're my friend here	But how are you going
Would you describe it as a friend	If you have to go to bed
He's my friend here	I don't want you to get back to work with us
But i'm a friend	What did you hear me
You have a good friend	And what do you have to be
And by the way down here and wait for her friend	Do you know what you're talking
Do you have a friend	Do you have to go to the
That's a good friend	And how are you sure it was just a little bit
He's my friend rachel	We're gonna need you to know about the
Talk to you as a friend of mine should be able	So how are you doing all this time
As my best friend and then when the time to	You got to go to the
I'm your best friend and a lot better than me	Do you have to go with the
What about our little friend over there to	Now why don't you tell them anything about this

(a) I'll talk to my friend and my sister

A(a) B(b) C(c) D(d) E(e) F(f) D(d) G(g)

(b) I'll talk to → AbbC + CbbA
talk to my → BccD + DccB
to my friend → CddE + EddC
my friend and → DeeF + Feed
friend and my → Efff + Dffe
and my sister → FddH + Hddf

(c) A = AAGTCATCGG a = CCGATGACTT
B = CAGTGCAGTG b = CACTGCACTG
C = ACTTCAAGGC c = GCCTTGAAGT
D = ATACCGACTG d = CAGTCGGTAT
E = TCGTCAGTCT e = AGACTGACGA
F = GTGTTGCACT f = AGTGCAACAC
H = ATCTTGGTGG h = CCACCAAGAT

Fig. 3. Encoding a sentence in DNA hyperedges. (a) Each word in the sentence is represented as a DNA sequence (A(a), B(b), ...), where the upper and lower cases indicate complementary sequences, respectively. (b) The word in the middle of a hyperedge of order 3 appears two times. The complementary DNA hyperedges (green dotted line) have a reverse sequence from the corresponding DNA hyperedge, playing a role of bridge that connects DNA hyperedges. (c) The DNA sequences of each word.



Fig. 4. Two forms of hybridization. (a) No repeating center DNA words. (b) Repeating center DNA words in DNA hyperedges.

green dotted lines in Fig. 3 are the reverse sequences of the hyperedges marked as black lines, playing a role of bridge between the two hyperedges of black lines. The black hyperedges also act as a bridge between the two green dotted hyperedges. Thus, the DNA molecules make context-appropriate sentences by repeating the center word and adding the reverse sequences.

2.3. Sequence design for DNA hyperedges

To avoid cross-hybridization between DNA hyperedges, the DNA sequences should be designed carefully. We minimized the possibility of the cross-hybridization by assigning sequences to the DNA hyperedges which show lowest free energies. Free energies between DNA hyperedges are estimated by Nearest-Neighbor thermodynamics (SantaLucia, 1998). The sequence optimization process consists of two steps: generation of DNA words and assigning DNA words to DNA hyperedges.

The first step is to generate a set of DNA words. Given a length of DNA word and a ratio of GC contents, a DNA word is randomly generated. The ratio of GC contents is the same for all DNA words to have similar melting temperature. We set the length of DNA word as 10 and the ratio of GC contents as 50% in the experiment. The DNA word which is equal to its reverse complement is discarded. If the half of a DNA word is a reverse complement of the other half, the DNA word and its reverse complement mean same word. For example, considering that a DNA word 'ACCTGCAGGT' is given, the front half 'ACCTG' is reverse complement of the other half 'CAGGT', the DNA word's reverse complement is also 'ACCTGCAGGT'. The generated DNA word is added to a pool of DNA words after comparing the similarity with the DNA words in the pool. If the similarity is higher than 80%, the DNA word is discarded. Finally, DNA sequences are generated for the required number of DNA words, and then, they are sorted in order of average free energy.

The second step of sequence optimization is to assign DNA words to the DNA hyperedges. Before the assignment step, the number of occurrences of each word in the hyperedges is counted. In order to assign better DNA words to the more frequently occurred words, the set of words are sorted by the occurrence frequency in decreasing order. The assignments are done in a greedy manner. From the pool of DNA words, the first DNA word is popped. The popped DNA word is assigned to the most frequently occurring word. All words in the DNA hyperedges are assigned in this manner. After filling out the words, the DNA words in the first DNA hyperedge are fixed. The other DNA hyperedges are computed free energies with the first DNA hyperedge. The free energy is computed by UNAFold (Markham and Zuker, 2008) which is a well-known program implementing Nearest-Neighbor thermodynamics. The DNA hyperedges which have the free energies over -20 kcal/mol have the possibility of cross-hybridization with the first DNA hyperedge. Therefore, we discard the DNA words assigned to the DNA hyperedges and replace them by regenerating DNA words by following the first step. If all the free energies are below -21 kcal/mol, it means that the first DNA hyperedge is safe against the cross-hybridization. Such an assignment step is performed for the second DNA hyperedge, and so on. Finally, the step is finished when all DNA hyperedges are fixed.

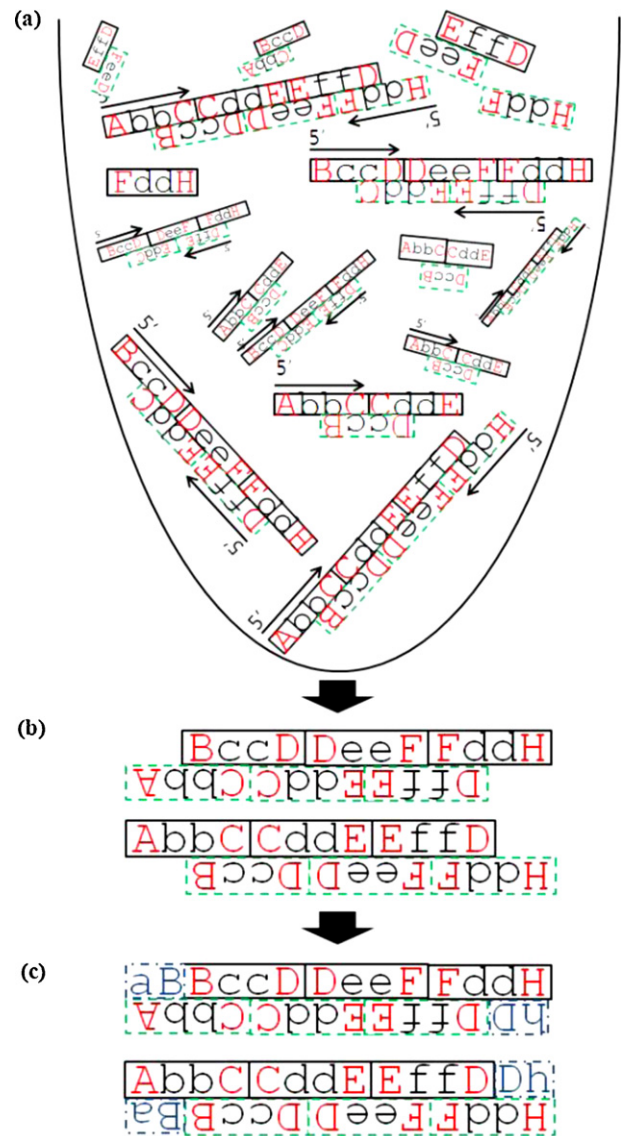


Fig. 5. Generating DNA sentences in a microtube. (a) The DNA hyperedges in the tube find their complementary pairs, assemble by hybridizing in a massively parallel way, and constitute DNA sentences. (b) An example DNA sentence being generated by DNA hyperedges. (c) The completed form of DNA sentences with the blue sequences denoting the end-filled DNA.

2.4. Generating sentences

Experimentally, the sentence generation is performed in a microtube in a hybridization buffer. As shown in Fig. 5, sentences are generated when dsDNAs are formed through hybridization of ssDNAs. This process is like performing one sweep of all the processes involved in sentence generation in the language hyper-network model. Given that there are twelve DNA hyperedges in the test tube, it is shown here that spontaneous self-assembly by the molecules generated DNA sentences. After the generation, we performed ligation of the 3' hydroxyl and the 5' phosphate using ligase to seal nicks. The DNA hyperedges are physically linked in this process, and the fixed DNA hyperedges are the final DNA sentences. The length of DNA sentences is determined by the number of linked DNA hyperedges. Since a DNA hyperedge is a 40-mer, a DNA sentence linked with three DNA hyperedges is a 120-mer. A sentence consisting of 4 hyperedges is a 160-mer. A DNA sentence linked with three DNA hyperedges contains 12 words in it. However, the generated sentences contain sticky ends which are 20-mer

single-strand overhang as shown in Fig. 5(b). Since these sentences are not complete double-strand DNA, we perform 1-cycle PCR to fill up the sticky ends as shown in Fig. 5 (c). This type of PCR method is only used to fill the single strand regions, and it does not amplify DNA.

2.5. Decoding

We perform sequencing to decode the generated DNA sentences. However, since the kinds and lengths of the generated sequences varied, we need an additional method in which a DNA of a specific length is sequenced. For that purpose, we use electrophoresis to arrange the generated DNA sentences by size, and then select (by gel) the sentences whose size is the same as that required. We extracted DNA of 140 bp to find the sentences generated as described in Section 2.4. Then, we performed T-vector cloning to isolate individual sentences since only one sentence can exist in a colony. After T-vector cloning, we choose sentences randomly by colony picking, and perform sequencing. In this process by performing more colony picking, we can find more sentences.

3. Experiments and results

To verify the feasibility of our DNA assembly model, we performed a restricted wet experiment. The nucleotide sequences for sentence generation are given in Table 2. All sequences were generated by the sequence design method as described in Section 2.3. Their free energy frequencies were checked as shown in Fig. 6.

All sequences were purchased from Bioneer (Daejeon, Korea), and each sequence was brought to a stock concentration 100 μ M in distilled water and stored at -20°C . The hybridization was performed in solution containing 300 mM NaCl using 12 sequences. The reaction mixture was incubated at 95°C for 3 min and the temperature was steadily lowered to 25°C by $1^{\circ}\text{C}/3$ min using a thermal cycler (iCycler, Bio-Rad). Ligations were performed for 1 h

Table 2
Designed DNA hyperedge sequences.

Hyperedges	DNA sequence
AbbC	AAGTCATCGGCACTGCACTGCACTGCACTGCACTTCAAGGC
BccD	CAGTGCAGTGGCCTTGAAGTGCCTTGAAGTATACCGACTG
CddE	ACTTCAAGGCCAGTCGGTATCAGTCGGTATTCGTCAGTCT
DeeF	ATACCGACTGAGACTGACGAAGACTGACGAGTGTGCACT
EffD	TCGTCAGTCTAGTGAACACAGTGAACACATACCGACTG
FddH	GTGTTGCACTCAGTCGGTATCAGTCGGTATATCTTGGTGG
CbbA	ACTTCAAGGCCAGTCGCACTGCACTGAAGTCATCGG
DccB	ATACCGACTGGCCTTGAAGTGCCTTGAAGTCAGTGCAGTG
EddC	TCGTCAGTCTCAGTCGGTATCAGTCGGTATATCTCAAGGC
FeeD	GTGTTGCACTAGACTGACGAAGACTGACGAATACCGACTG
Dffe	ATACCGACTGAGTGAACACAGTGAACACTCGTCAGTCT
HddF	ATCTTGGTGGCAGTCGGTATCAGTCGGTATGTGTTGCACT

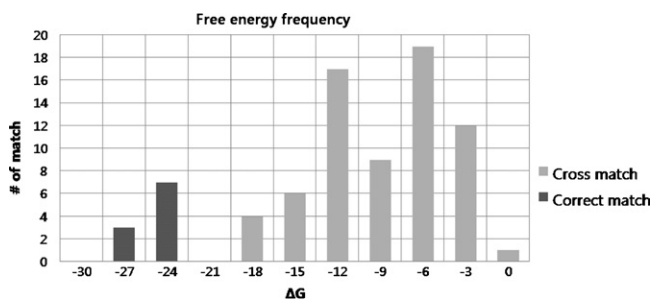


Fig. 6. The free energy diagram of the number of matches in the DNA sequences designed. The black region (left part) shows 10 correct matches which have complement sequences with each other. The gray region (right part) shows 68 cross matches which have partial complementary sequences with each other.

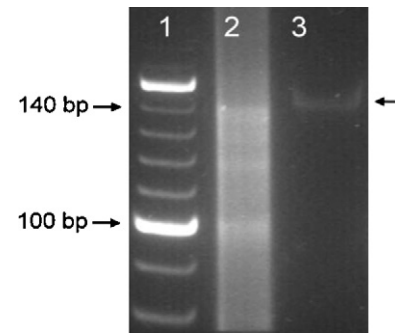


Fig. 7. Gel electrophoresis image. Lane 1 denotes 10-bp markers. Lane 2 shows smearing because the generated sentences and hyperedges are mixed. Lane 3 denotes the end filled DNA sentences (140-mer) linked with three hyperedges. The difference between the number of matching base pairs and 140 bp is from sequencing errors.

at room temperature in a final volume of 10 μ l with $1\times$ quick ligation reaction buffer, 5 μ l hybridized DNA, and 1 μ l quick T4 DNA ligase (Quick ligation kit, New England Biolabs). Then to complete the sentence, the ligated DNA was used in an end filling reaction with Taq DNA polymerase (AccuPower PCR premix, Bioneer) for 5 h at 68°C . End filled fragments were separated on a 12% polyacrylamide gel to find three hyperedges linked DNA sentences. After electrophoresis, the gel was stained with SYBR Gold (Invitrogen/Molecular Probes) in TBE buffer for 40 min. The gel was visualized under UV light, and the image was captured by UV-Gel Doc system (Bio-Rad) as shown in Fig. 7. The several lengths of generated sentences are divided in the gel (lane 2) in Fig. 7. For separating DNA, it was extracted from the gel using standard phenol/chloroform extractions and purified by ethanol precipitation. The separated DNA sentences are shown in lane 3 in Fig. 7. Subsequently, the extracted DNA was cloned by pGEM-T Easy Vector System (Promega). In the cloning process, transformation was conducted by electroporation. We used ElectroMax DH10B (Invitrogen) competent cell and MicroPulser (Bio-Rad). Then we selected colonies and performed sequencing with 3730 DNA analyzer (Applied Biosystems). We extracted DNA from 40 colonies to get cloned DNA sentences. Nine of them are removed because three did not have an inserted DNA sentence in the vector and six had failed decoding. It was verified that in the sentences, about 90% (126-mer) were equal to AbbCCddEEffDDh ("I'll talk to my friend and my sister") or 25 in the total of 31 DNA sentences, as shown in Table 3. In other words, we confirmed

Table 3

Sequencing results of the generated sentence ("I'll talk to my friend and my sister", AbbCCddEEffDDh). The number of base matches to this sentence are in the range to 140–85-bp. The plus and minus signs mean the sequence direction.

Match (bp)	Strand	Sentence	Match (bp)	Strand	Sentence
119	+	Nonsense	130	–	AbbCCddEEffDDh
139	+	AbbCCddEEffDDh	138	–	AbbCCddEEffDDh
139	+	AbbCCddEEffDDh	125	–	Nonsense
140	+	AbbCCddEEffDDh	85	–	Nonsense
118	+	Nonsense	140	–	AbbCCddEEffDDh
140	+	AbbCCddEEffDDh	140	–	AbbCCddEEffDDh
139	+	AbbCCddEEffDDh	139	–	AbbCCddEEffDDh
129	+	AbbCCddEEffDDh	91	–	Nonsense
140	+	AbbCCddEEffDDh	140	–	AbbCCddEEffDDh
132	–	AbbCCddEEffDDh	136	–	AbbCCddEEffDDh
113	–	Nonsense	137	–	AbbCCddEEffDDh
140	–	AbbCCddEEffDDh	140	–	AbbCCddEEffDDh
138	–	AbbCCddEEffDDh	140	–	AbbCCddEEffDDh
139	–	AbbCCddEEffDDh	137	–	AbbCCddEEffDDh
133	–	AbbCCddEEffDDh	132	–	AbbCCddEEffDDh
140	–	AbbCCddEEffDDh			

that the decoding ratio, i.e. how many of the DNA sentences are decoded correctly from the cloned sequences, is 80% (20% are nonsense).

4. Concluding remarks

We have designed a DNA encoding method for words and phrases, and presented a DNA assembly model for sentence generation which is based on the hypernetwork structure. We performed *in vitro* DNA experiments and demonstrated that context-appropriate sentences are generated from the DNA language model. Since the DNA language model can be constructed from a large corpus of sentences, the DNA assembly model can be used to simulate usage-based linguistics and corpus-based language processing. In particular, the parallel and associative constraint satisfaction processes underlying the DNA self-assembly show the usefulness of molecular language processing to implement and simulate the statistical language models such as the maximum entropy model in a nonconventional way. This proof-of-principle was given by *in silico* simulation of sentence generation from a large corpus of *Friends* sentences. The examples of generated sentences show that the DNA assembly model constructs stochastically the sentences that best match with the given query words and the words generated in the previous steps. One drawback in DNA implementation of language generation using the current technology is its cost. Oligomer synthesis and DNA sequencing are still very expensive, and we were forced to experiment in this paper with a small vocabulary size. However, these hurdles are expected to be overcome with the recent rapid pace of technology development for DNA writing and reading. Recent advance of high-throughput sequencing technology, such as Solexa, 454, and SOLiD, can provide a solution to these problems by sequencing 10^6 DNA sequences at once (Shendure and Ji, 2008).

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References

- Adleman, L.M., 1994. Molecular computation of solutions to combinatorial problems. *Science* 266, 1021–1024.
- Biber D., Conrad S., Reppen R., 1998. *Corpus Linguistics*. Cambridge Univ. Press.
- Foster, M., 2007. Issues for corpus-based multimodal generation. *CiteSeer*, 51–58.
- Heylen, K., Tummer, J., Geeraerts, D., 2008. Methodological issues in corpus-based cognitive linguistics. *Cognitive Sociolinguistics: Language Variation Cultural Models, Social Systems*, 91–128.
- Lee, J.-H., Lee, E.S., Zhang, B.-T., 2009. Hypernetwork memory-based model for infant's language learning. *Journal of KIIE: Computing Practices and Letters* 15, 983–987.
- Mao, C., LaBean, T.H., Reif, J.H., Seeman, N.C., 2000. Logical computation using algorithmic self-assembly of DNA triple-crossover molecules. *Nature* 407, 493–496.
- Markham, N.R., Zuker, M., 2008. UNAFold: software for nucleic acid folding and hybridization. *Methods in Molecular Biology* 453, 3–31.
- Maune, H.T., Han, S.P., Barish, R.D., Bockrath, M., Iii, W.A., Rothmund, P.W., Winfree, E., 2010. Self-assembly of carbon nanotubes into two-dimensional geometries using DNA origami templates. *Nature Nanotechnology* 5, 61–66.
- McEnery, T., Xiao, R., Tono, Y., 2006. *Corpus-based Language Studies: An Advanced Resource Book*. Taylor & Francis.
- Paun, G., Rozenberg, G., Salomaa, A., 1998. *DNA Computing: New Computing Paradigms*. Springer-Verlag.
- Reif, J.H., LaBean, T.H., 2007. Autonomous programmable biomolecular devices using self-assembled DNA nanostructures. *Communications of the ACM* 50 (9), 46–53.
- Rosenfeld, R., Chen, S.F., Zhu, X., 2001. Whole-sentence exponential language models: a vehicle for linguistic-statistical integration. *Computer Speech and Language* 15, 55–73.
- SantaLucia Jr., J., 1998. A unified view of polymer, dumbbell, and oligonucleotide DNA nearest-neighbor thermodynamics. *Proceedings of the National Academy of Sciences of the United States of America* 95, 1460–1465.
- Seeman, N.C., 2003. DNA in a material world. *Nature* 421, 427–431.
- Shendure, J., Ji, H., 2008. Next-generation DNA sequencing. *Nature Biotechnology* 26, 1135–1145.
- Tomasello, M., 2003. *Constructing a Language: A Usage-based Theory of Language Acquisition*. Harvard University Press.
- Tummers, J., Heylen, K., Geeraerts, D., 2005. Usage-based approaches in cognitive linguistics: a technical state of the art. *Corpus Linguistics and Linguistic Theory* 1, 225–261.
- Yan, H., LaBean, T.H., Feng, L., Reif, J.H., 2003. Directed nucleation assembly of bar-code patterned DNA lattices. *Proceedings of the National Academy of Sciences* 100 (14), 8103–8108.
- Winfree, E., Liu, F., Wenzler, L.A., Seeman, N.C., 1998. Design and self-assembly of two-dimensional DNA crystals. *Nature* 394, 539–544.
- Zhang, B.-T., 2008. Hypernetworks: a molecular evolutionary architecture for cognitive learning and memory. *IEEE Computational Intelligence Magazine* 3, 49–63.
- Zhang, B.-T., Kim, J.-K., 2006. DNA hypernetworks for information storage and retrieval. *DNA* 12, 283–292.
- Zhang, B.-T., Park, C.H., 2008. Self-assembling hypernetworks for cognitive learning of linguistic memory. *Proceedings of the World Academy of Science and Engineering Technology (WASET)*, 134–138.
- Zheng, J., Birktoft, J.J., Chen, Y., Wang, T., Sha, R., Constantinou, P.E., Ginell, S.L., Mao, C., Seeman, N.C., 2009. From molecular to macroscopic via the rational design of a self-assembled 3D DNA crystal. *Nature* 461, 74–77.